FORENSIC COMPLIANCE & LEGAL ADMISSIBILITY REPORT

Malkhana Vault Compliance with BSA, 2023 and BNSS, 2023 Standards

Malkhana Vault is built specifically to address the digital evidence custody requirements of the **Bharatiya Sakshya Adhiniyam (BSA), 2023** and the **Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023**. This document outlines how the application's technical architecture maps directly to the legal requirements for presenting digital evidence in Indian courts.

1. Compliance Mapping Table

Legal Provision	Statutory Requirement	System Safeguard / Implementation
BSA §61	Admissibility of Electronic Records	Establishes the electronic record's baseline identity through device fingerprinting (MAC, OS version, application build) and temporal Indian Standard Time (IST) locks.
BSA §63	Admissibility Certificate Requirement	Automates the generation of the mandatory court admissibility certificate, eliminating drafting errors. Includes digital signatures of both the Investigating Officer (IO) and the Forensic Specialist.
BSA §63(2)(c)	System Operational Integrity	Background daemon logs all **`STARTUP`**, **`SHUTDOWN`**, **`POWER_LOSS`**, and **`CRASH`** events. The certificate engine scans these logs and automatically appends a downtime disclosure if any interruptions occurred during custody.
BNSS §153 & §173	Mandatory Crime Scene Digital Seizure	Mandates the immediate capture of Panch (witness) names, seizure coordinates, and the generation of the **H1 Birth Hash** directly at the scene of the crime before transit.
IT Act §7A	Secure Audit Trails (180 days+)	Keeps a cryptographically sealed, sequential, chronological audit log. Each event is bound using SHA-256 in a Merkle log chain where deletion or modification is impossible.

2. Technical Compliance Proofs

2.1 The Triple-Hash Protocol (No "Malkhana Gap")

To prove that digital evidence was not tampered with during transit from the crime scene to the station malkhana, or during extraction at the Forensic Science Laboratory (FSL), the system enforces three cryptographic checkpoints:

1. **H1 (Birth Hash):** Generated immediately upon seizure in the field.
2. **H2 (Receipt Hash):** Generated by the Malkhana In-Charge before placing the device in the physical coordinate drawer grid. The system blocks normal intake if $H_1 \neq H_2$, alerting the operator to a "Malkhana Gap."
3. **H3 (Analysis Hash):** Generated by the FSL examiner post-imaging, proving that the forensic extraction process did not alter the source media.

2.2 System Downtime Mechanics (BSA §63(2)(c))

A common challenge in prosecuting digital evidence is proving that the computer system storing or processing the data was operating properly.

- **WAL Journaling:** Malkhana Vault utilizes SQLite in Write-Ahead Logging (WAL) mode. Transactions are committed to a roll-forward log file, protecting the database from corruption during unexpected power outages.
- **Automated Attestation:** If a power failure or system crash occurs, the event is logged. The generated Section 63 certificate automatically includes these logs, disclosing to the court that the database's cryptographic integrity was verified and maintained through WAL recovery on startup.

2.3 Zero-Trust, Append-Only Storage

To satisfy the court that records have not been altered or deleted by police operators:

- **No Delete Queries:** The codebase contains no SQL `DELETE` or `UPDATE` queries for evidence or audit logs.
- **SQLCipher AES-256:** The database file is encrypted at rest using AES-256-CBC with 256,000 PBKDF2 iterations.
- **Merkle Chain Verification:** Every event (logins, evidence views, transfers) is hashed sequentially:

$$H_{\text{entry}} = \text{SHA-256}(H_{\text{prev}} \mathbin{\Vert} \text{Type}_{\text{event}} \mathbin{\Vert} \text{Actor} \mathbin{\Vert} \text{Timestamp})$$

Any manual manipulation of the database file breaks this hash chain, causing audit validation to fail immediately.